**QPCODE:** **Reg. No : .....**

**Name : .....**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**FIFTH SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION**
**Fifth Semester**

**MG5DSECFS308 SECURITY THREATS AND VULNERABILITIES**

**(2024 ADMISSION ONWARDS)**

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Explain the term insider? [U] / [1]
2. Identify four major physical threats to information systems. [U] / [1]
3. Define "social engineering" in the context of e-mail security and mass mailer viruses. [K] / [1]
4. List the four major areas of client-side security. [K] / [2]
5. Classify the three basic methods of user identification/authentication. [A] / [2]
6. Identify any two authentication services used on the Internet. [U] / [2]
7. Classify IDSs based on their sources of audit information. [A] / [3]
8. Name two examples of network-based intrusion detection systems. [K] / [3]
9. Classify cyber attacks into passive attack, active attack, internal misuse, and external misuse. [A] / [3]
10. Compare external auditors and internal auditors with respect to their roles. [U] / [4]

11. Select any two detective techniques that can be used to identify information leakage. [E] / [4]

12. Evaluate the difference between Rights Management and Rights Enforcement. [E] / [4]

**[10x2 = 20]**

## Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Compare infrastructure mode and ad hoc mode in wireless networks with suitable examples. [U] / [1]

14. Examine buffer overflow vulnerabilities, their effects, and countermeasures. [An] / [2]

15. Describe the process of developing and distributing a virus signature. [U] / [2]

16. Classify NIDS detection methods into anomaly-based and misuse-based detection. [A] / [3]

17. Define security awareness and identify the major benefits of security-aware employees. [K] / [3]

18. Evaluate the four major Web-hosting categories: shared servers, dedicated servers, colocation, and managed services. [E] / [4]

19. Compare Dedicated mode, System High mode, and Multilevel mode with respect to users, access control, and security requirements. [An]/[4]

20. Classify the DoD security levels into Top Secret, Secret, Confidential, and Unclassified, and explain the access rights associated with each level. [A] / [4]

**[6x5 = 30]**

## Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks.

21. Classify hacking toolkits based on their procedural and functional classifications. Explain the purpose of each category with suitable examples. [U / [1]

22. Differentiate between tamper-evident, tamper-resistant, and tamper-reactive security mechanisms used in HSMs, with suitable examples. [An] / [2]

23. Classify different types of security incidents and explain the appropriate response for each type. [A] / [3]

**[2x10 = 20]**